The tempered-likelihood learner: definition and agnostic guarantee

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This is a self-contained statement of the main positive result of the agnostic-learning note (`neural-network-generalization/works-in-progress/agnostic-learning/`). Theorem 1 and Corollary 2 are Theorem 6.3 and Corollary 8.1 of `research-notes/B-tempered-likelihood.md`, cited below as B, stated here without proofs. Section 5 also summarizes results of B and of `E-finite-prior-scale.md` (E) and `A-tight-conditioning-fails.md` (A) in the same folder. Their proofs are paper proofs with numerical checks, refereed by independent AI agents; they are not Lean-verified. Corollary 3 in Section 4 and the remark "One tempering for a range of class sizes" in Section 3 are derived here from Theorem 1 and Corollary 2, with their short arguments, and have not been refereed.

1. Setting

Data. Fix $d \geq 1$ and a nonempty compact set $X \subseteq [-1,1]^d$. A *data law* D is any Borel probability measure on $X \times [-1,1]$. Nothing else is assumed: within $[-1,1]$, the labels may be noisy, or any deterministic or random function of the input. Unbounded label noise, such as Gaussian noise added to a target, is not covered. The sample $S = ((x_i, y_i))_{i=1}^n$ consists of n independent draws from D. For $f : X \to \mathbb{R}$, the test loss and the training loss are

$$L_D(f) = \mathbb{E}_{(x,y)\sim D}\,(f(x)-y)^2, \qquad \hat{L}_S(f) = \frac{1}{n}\sum_{i=1}^n \big(f(x_i)-y_i\big)^2,$$

and $\mathrm{clip}(u) = \max\{-1, \min\{1, u\}\}$ truncates a real number to the label range.

Networks. The depth is $L = H+1 \geq 2$, with H hidden layers, and the width is N. A parameter is $w = (W^1, \dots, W^H, v)$, with $W^1 \in \mathbb{R}^{N\times(d+1)}$, $W^\ell \in \mathbb{R}^{N\times N}$ for $2 \leq \ell \leq H$, and $v \in \mathbb{R}^N$. With $\bar{x} = (1, x)$ and tanh applied entrywise, the network computes

$$h^0(x) = \bar{x}, \qquad h^\ell(x) = \tanh\big(W^\ell h^{\ell-1}(x)\big)\ \ (1 \leq \ell \leq H), \qquad z_w(x) = v^\top h^H(x).$$

There are no biases other than the constant coordinate of $\bar{x}$. The *energy* of w is

$$\kappa_N(w) = \frac{d}{N}\,\|W^1\|_F^2 + \sum_{\ell=2}^{H} \|W^\ell\|_F^2 + N\,\|v\|_2^2.$$

Definition (the tempered-likelihood learner). Fix the depth L, a learner width N, a scale $\sigma > 0$ and an exponent $\theta \in (0,1)$, and put $c = n^{1-\theta}$.

- *Prior.* $\mu_\sigma(dw) \propto e^{-\kappa_N(w)/(2\sigma^2)}\,dw$: independent centred Gaussian weights with variances $N\sigma^2/d$ (first layer), σ^2 (internal layers) and σ^2/N (readout).
- *Likelihood.* $y_i \mid x_i, w \sim \mathrm{N}\big(z_w(x_i),\, c\,\sigma^2\big)$: Gaussian observation noise whose variance is the prior scale σ^2 times the tempering factor c.
- *Posterior.*

$$\Pi_\sigma(dw) \ \propto\ e^{-F_{c,S}(w)/(2\sigma^2)}\,dw, \qquad F_{c,S}(w) = \kappa_N(w) + \frac{1}{c}\sum_{i=1}^n \big(z_w(x_i)-y_i\big)^2 = \kappa_N(w) + n^\theta\,\hat{L}_S(z_w).$$

- *Strong-prior limit.* The sample size n, the sample S and the width N are fixed first; then $\sigma \downarrow 0$.

The learner's choices are the depth L, the width N and the exponent θ; none of them depends on the data law, and the guarantees below hold at every width at least the competitor's. With $c = 1$, that is without tempering, this is the plain strong-prior Bayesian posterior with a Gaussian likelihood. The tempering $c = n^{1-\theta}$ raises its likelihood to the power $n^{\theta-1}$, so the learner is a power posterior whose temperature depends on n, not one Bayesian model updated as data arrive.

Predictions. As $\sigma \downarrow 0$ the posterior concentrates on the *near-minimizers*

$$\mathcal{N}_1 = \{w : \ F_{c,S}(w) \le \min F_{c,S} + 1\}$$

(the minimum exists). Equivalently, $\mathcal{N}_1$ is the set where the energy-penalized least-squares objective $\hat{L}_S(z_w) + n^{-\theta}\kappa_N(w) = n^{-\theta}F_{c,S}(w)$ is within $n^{-\theta}$ of its minimum. The guarantee covers three ways of predicting, all with outputs clipped to $[-1,1]$:

1. a near-minimizer: $\operatorname{clip} z_w$ for any $w \in \mathcal{N}_1$;
2. a posterior draw, through its expected test loss $\mathbb{E}_{w\sim\Pi_\sigma} L_D(\operatorname{clip} z_w)$ (the clipped Gibbs risk);
3. the posterior mean of the clipped outputs, $\bar{f}_\sigma(x) = \mathbb{E}_{w\sim\Pi_\sigma} \operatorname{clip} z_w(x)$.

The likelihood uses the unclipped output z_w; clipping is applied only when predicting. Since every label lies in $[-1,1]$, clipping never increases a squared error.

Competitors. A competitor is any network $g = z_{w^g}$ of the same architecture, with depth L, width m and all weights in $[-B,B]$, $B > 0$, and we put

$$E_* = d(d+1)B^2 + 2(L-1)\,m^2B^2, \qquad U = mB, \qquad \Gamma_n = \sqrt{2(d+1)\ln(n+1)} + \frac{H-1}{2}$$

(Γ_n is written D_n in B). For every width $N \ge m$, some width-N network computing exactly g (copy each hidden unit of g onto a group of $\lfloor N/m \rfloor$ or $\lceil N/m \rceil$ units and divide its outgoing weights by the group size) has energy at most E_*, and $U \ge \sup_x |g(x)|$.

2. The guarantee

> **Theorem 1 (agnostic oracle inequality at every learner width; B, Theorem 6.3).** Let $L = H+1 \ge 2$, $\theta \in (0, 1/H)$ and $c = n^{1-\theta}$. Let $n \ge 1$, X, the data law D, the competitor g (width m, weights in $[-B,B]$) and $\delta \in (0,1)$ be arbitrary, and put
>
> $$\mathrm{Exc}_n = (E_*+1)\,n^{-\theta} + 8\Big(\frac{2}{H}\Big)^{H/2}\Gamma_n\,n^{-(1-\theta H)/2} + \big(4 + (U+1)^2\big)\sqrt{\frac{\ln(2/\delta)}{2n}}.$$
>
> There is an event of probability at least $1-\delta$ over the sample, which depends on g but not on the learner width or on σ, on which, for **every** learner width $N \ge m$:
>
> a. every near-minimizer $w \in \mathcal{N}_1$ has $L_D(\operatorname{clip} z_w) \le L_D(g) + \mathrm{Exc}_n$;
> b. $\Pi_\sigma\{w : \ L_D(\operatorname{clip} z_w) > L_D(g) + \mathrm{Exc}_n\} \to 0$ as $\sigma \downarrow 0$;
> c. $\limsup_{\sigma\downarrow 0}\ \mathbb{E}_{w\sim\Pi_\sigma} L_D(\operatorname{clip} z_w) \le L_D(g) + \mathrm{Exc}_n$;
> d. $\limsup_{\sigma\downarrow 0}\ L_D(\bar{f}_\sigma) \le L_D(g) + \mathrm{Exc}_n$.
>
> Moreover, for every $\varepsilon > 0$, $\mathrm{Exc}_n \le \varepsilon$ whenever $n \ge n_0$, where
>
> $$n_0 := \max\Big\{\Big(\frac{3(E_*+1)}{\varepsilon}\Big)^{\frac{1}{\theta}},\ \Big(\frac{48(2/H)^{H/2}}{\varepsilon}\sqrt{\frac{4(d+1)}{1-\theta H}}\Big)^{\frac{4}{1-\theta H}},\ \Big(\frac{24(2/H)^{H/2}(H-1)}{\varepsilon}\Big)^{\frac{2}{1-\theta H}},\ \frac{9\big(4+(U+1)^2\big)^2\ln(2/\delta)}{2\varepsilon^2}\Big\},$$
>
> and the third entry is omitted when $H = 1$. So the sample size is polynomial in d, m, B, $1/\varepsilon$ and $\log(1/\delta)$ at fixed L and θ.

In words: whatever the data law, after n samples, with probability at least $1-\delta$ and in the limit $\sigma \downarrow 0$, the tempered posterior at every width at least the competitor's predicts (after clipping) at least as well as the fixed network g, up to an error Exc_n. The error tends to 0 as $n \to \infty$, grows with the competitor's width and weight bound, and does not depend on the learner width. For example, $\theta = 1/(H+2)$ gives

$$\mathrm{Exc}_n \le \big[E_* + 1 + 8(2/H)^{H/2}\Gamma_n\big]\,n^{-1/(H+2)} + \big(4+(U+1)^2\big)\sqrt{\frac{\ln(2/\delta)}{2n}} = \tilde{O}\big(\mathrm{poly}(d,m,B)\,n^{-1/(H+2)}\big).$$

Section 4 makes the polynomial explicit. The constants are poor: at $L=3$, $d=10$, $m=B=1$, $\varepsilon=0.1$, $\delta=0.05$ and $\theta=1/4$, Exc_n first drops below ε at $n \approx 10^{14.1}$, and the closed-form n_0 is about $10^{29.2}$.

3. Against networks whose width and weights are at most n^α

Corollary 2 (B, Corollary 8.1). Fix $L = H + 1 \geq 2$ and $\alpha \in \big(0, \frac{1}{4H}\big)$. Put $\theta = \frac{1+8\alpha}{H+2} \in (0, \frac{1}{H})$ and $c = n^{1-\theta}$. Let $\mathcal{G}_n^{(\alpha)}$ be the networks of the architecture above with depth L, width at most $\lfloor n^\alpha \rfloor$ and all weights in $[-n^\alpha, n^\alpha]$, and put

$$\beta = \frac{1 - 4\alpha H}{H+2} \ \ (H \geq 2), \qquad \beta = \min\Big\{\frac{1-4\alpha}{3},\ \frac{1}{2} - 4\alpha\Big\} \ \ (H = 1),$$

$$\Xi_n = d(d+1) + 2L + 8\Big(\frac{2}{H}\Big)^{H/2}\Gamma_n + 4\sqrt{2\ln(2/\delta)}.$$

Then for every d, every compact $X \subseteq [-1,1]^d$, every data law D on $X \times [-1,1]$, every $n \geq 1$ and every $\delta \in (0,1)$, with probability at least $1 - \delta$, simultaneously for all learner widths $N \geq n^\alpha$,

$$\limsup_{\sigma \downarrow 0}\ \mathbb{E}_{w \sim \Pi_\sigma} L_D(\operatorname{clip} z_w) \ \leq\ \min_{f \in \mathcal{G}_n^{(\alpha)}} L_D(f) + \Xi_n\, n^{-\beta},$$

and the same bound holds for every near-minimizer and for the posterior mean of the clipped outputs $\bar{f}_\sigma$; the ordinary posterior mean $\mathbb{E}_{\Pi_\sigma} z_w$ is not covered. When $H = 1$, $\beta > 0$ only for $\alpha < 1/8$.

For $H \geq 2$ this has the form of the motivating conjecture (test loss at most the least test loss over networks of size $\mathrm{poly}(n)$, plus a vanishing term), restricted to small exponents and bounded weights: the clipped test loss is at most the least test loss over depth-L networks of the architecture above (tanh, no biases beyond the constant input) whose width and weights are at most n^α, plus $O_L\big((d^2 + \sqrt{d \log n} + \sqrt{\log(1/\delta)})\, n^{-\beta}\big)$, for every $\alpha < 1/(4H)$; at depth 3 one may take $\alpha = 1/16$, $\beta = 1/8$. It does not cover all polynomial sizes: the weights must be bounded as well as the width, and Section 5 reports (B, Corollary 9.3) that, once $\alpha > L/(2H(2L-1))$, no tempering $c(n)$ gives a polynomial sample bound for the clipped predictions that is uniform over laws and input dimensions and holds at every learner width $N \geq n^\alpha$.

One tempering for a range of class sizes (derived here from Theorem 1, by the arithmetic of the proof of B's Corollary 8.1; not stated or refereed in the notes). Let $H \geq 2$ and fix any $\theta \in (0, 1/H)$, without reference to α. For $\alpha \in (0, \theta/4)$, apply Theorem 1 with $m = \lfloor n^\alpha \rfloor$ and $B = n^\alpha$ to a minimizer g of L_D over $\mathcal{G}_n^{(\alpha)}$; it exists because the parameter set is compact once narrower networks are padded with zeros, and it depends only on D, n and α, not on the sample. Since $E_* + 1 \leq (d(d+1) + 2L)\, n^{4\alpha}$ and $4 + (U+1)^2 \leq 8n^{4\alpha}$, there is an event of probability at least $1 - \delta$, which depends on α, on which statements (a)–(d) of Theorem 1 hold at every learner width $N \geq n^\alpha$, with $L_D(g)$ replaced by $\min_{f \in \mathcal{G}_n^{(\alpha)}} L_D(f)$ and Exc_n replaced by

$$(d(d+1) + 2L)\, n^{-(\theta - 4\alpha)} + 8\Big(\frac{2}{H}\Big)^{H/2}\Gamma_n\, n^{-(1-\theta H)/2} + 4\sqrt{2\ln(2/\delta)}\ n^{-(1/2 - 4\alpha)}.$$

This is $\tilde{O}\big(n^{-\min\{\theta - 4\alpha,\, (1-\theta H)/2\}}\big)$, since the third exponent never binds when $H \geq 2$. So one θ, chosen without knowing α, gives a vanishing bound for each fixed $\alpha \in (0, \theta/4)$, and θ close to $1/H$ covers every α up to nearly $1/(4H)$. The rate is never faster than the rate $n^{-\beta}$ of Corollary 2 and equals it only at $\alpha = ((H+2)\theta - 1)/8$. No claim is made simultaneously over α.

4. The error terms in polynomial form

Corollary 3 (derived here from Theorem 1 and Corollary 2; not refereed). Let $L = H + 1 \geq 2$, $d \geq 1$, $n \geq 1$ and $\delta \in (0,1)$.

i. In Theorem 1, for every $\theta \in (0, 1/H)$, with $r = \min\{\theta,\ (1-\theta H)/2\}$,

$$\mathrm{Exc}_n \ \leq\ 11\, L\,(B+1)^2 (d+m+1)^2 \sqrt{\ln(2/\delta)}\ \cdot\ \sqrt{\ln(n+1)}\, n^{-r}.$$

The choice $\theta = 1/(H+2)$ gives $r = 1/(H+2)$.

ii. In Corollary 2,

$$\Xi_n \ \leq\ 8\, L\,(d+1)^2 \sqrt{\ln(2/\delta)}\ \cdot\ \sqrt{\ln(n+1)}.$$

So, with probability at least $1-\delta$: statements (a)–(d) of Theorem 1 with $\theta = 1/(H+2)$ hold at every learner width $N \geq m$ with Exc_n replaced by

$$\mathrm{poly}\big(L,\, d,\, m,\, B,\, \log(1/\delta)\big) \cdot \sqrt{\log(n+1)} \cdot n^{-1/(H+2)},$$

and the conclusions of Corollary 2 hold at every learner width $N \geq n^{\alpha}$ with $\Xi_n\, n^{-\beta}$ replaced by

$$\mathrm{poly}\big(L,\, d,\, \log(1/\delta)\big) \cdot \sqrt{\log(n+1)} \cdot n^{-\beta}.$$

The prefactors in (i) and (ii) serve as the polynomials: they have degree 1 in L, 2 in each of d, m and B, and $\frac{1}{2}$ in $\ln(2/\delta)$. In the second, the competitors' size n^{α} enters only through the exponent β.

Proof. Put $a = \sqrt{\ln(n+1)}$ and $b = \sqrt{\ln(2/\delta)}$. Since $n \geq 1$ and $\delta < 1$, $a, b \geq \sqrt{\ln 2}$, so $1 \leq ab/\ln 2$ and $a, b \leq ab/\sqrt{\ln 2}$. Also $(2/H)^{H/2} \leq \sqrt{2}$ for every $H \geq 1$, since the left side decreases in H. (i) Put $Q = (B+1)(d+m+1) \geq 2$. Then

$$E_* + 1 \leq (2L-1)B^2(d+m+1)^2 + 1 \leq 2LQ^2, \quad 4 + (U+1)^2 \leq 4 + Q^2 \leq 2Q^2, \quad \sqrt{2(d+1)} \leq \sqrt{2}\, Q, \quad \frac{H-1}{2} \leq \frac{L}{2},$$

and each of $n^{-\theta}$, $n^{-(1-\theta H)/2}$ and $n^{-1/2}$ is at most n^{-r}, because $r \leq \theta$ and $r \leq (1-\theta H)/2 \leq 1/2$. Hence

$$n^{r}\, \mathrm{Exc}_n \leq 2LQ^2 + 16Qa + 4\sqrt{2}L + \sqrt{2}Q^2 b.$$

Using $Q \geq 2$ and $L \geq 2$, the four terms are at most 2.89, 4.81, 2.05 and 0.85 times LQ^2ab, a total below 11. (ii) In the same way,

$$\Xi_n \leq (d+1)^2 + 2L + 16\sqrt{d+1}\, a + 4\sqrt{2}L + 4\sqrt{2}b,$$

and using $\sqrt{d+1} \leq (d+1)^2/2^{3/2}$, $(d+1)^2 \geq 4$ and $L \geq 2$, the five terms are at most 0.73, 0.73, 3.40, 2.05 and 0.85 times $L(d+1)^2 ab$, a total below 8. □

The constants are not optimized, but they cannot be lowered below $2/\ln 2 \approx 2.89$ in (i) (let $L, m, B \to \infty$ with $d = n = 1$ and $\delta \to 1$) or below 5.32 in (ii) (take $d = n = 1$, $L = 2$ and $\delta \to 1$).

5. What else is known (scope)

- **Unclipped predictions** (B, Theorem 6.4). The same conclusions hold for z_w, for the expected test loss $\mathbb{E}_{\Pi_\sigma} L_D(z_w)$ of a posterior draw, and for the ordinary posterior mean $\bar{z}_\sigma = \mathbb{E}_{\Pi_\sigma} z_w$ and its clip, provided $\theta \in (0, 1/L)$, with a larger error term. For $\theta \in [1/L, 1/H)$ the unclipped predictors are open.
- **Sharpness** (B, Theorem 7.4). For $H \geq 2$ and $\theta > 1/H$, the learner memorizes label noise. Take inputs uniform on $\{-1,1\}^d$, fair random labels independent of the input, and $n = 2^d$ samples; the zero network has test loss 1, the Bayes risk. With probability at least $1 - e^{-n^{1/3}/2}$, at every learner width $N \geq N_0(n) = \lceil 2(d+1)^3 n^{1+\theta} \ln 2 \rceil + 1$, every near-minimizer has test loss at least $1 + 1/e - o(1)$, clipped or not, and so do the limits inferior as $\sigma \downarrow 0$ of the expected test loss of a posterior draw (clipped or not) and of the test losses of the posterior means $\bar{f}_\sigma$, $\bar{z}_\sigma$ and clip $\bar{z}_\sigma$. The exponent $1/H$ is therefore sharp in the worst case over data laws. The bound is asymptotic (for $H = 2$ and $\theta = 0.55$ it is 0 at $d = 50$, 100 and 200), the counterexample needs the input dimension $d = \log_2 n$ to grow, and at a fixed d, or at learner widths below $N_0(n)$, nothing is shown.
- **Competitor weights** (B, Section 9). For this learner, with any tempering $c(n)$ in place of $n^{1-\theta}$, bounded competitor weights are needed for a guarantee uniform over laws. For $H \geq 2$ and the classes $\mathcal{G}_n^{(\alpha)}$, once

$\alpha > L/(2H(2L-1))$ no $c(n)$ gives a polynomial sample bound for the clipped predictions that is uniform over laws and input dimensions and holds at every learner width $N \geq n^{\alpha}$; the failing laws change with n (the noisy cube in dimension $\log_2 n$ at large learner widths, or a two-point law in dimension 1 with inputs $\pm\delta_0 \to 0$). For a fixed data law and $\theta \in (0, 1/H)$ no weight bound is needed: by the universal approximation theorem, with probability at least $1-\delta$, the clipped test loss comes within any $\varepsilon > 0$ of the Bayes risk once n and the learner width exceed thresholds that depend on the law (B, Remark 9.4).

- **Other competitors** (B, Remark 8.2). A depth-L tanh network with biases, width m, and weights and biases in $[-B, B]$ equals a network of the architecture above with width $m+1$ and weights in $[-B', B']$, $B' = \max\{B, 1\}/\tanh 1$; Theorem 1 applies to it with these values. Competitors of smaller depth or with other activations (such as ReLU) are not covered.
- **The prior scale.** Everything above is stated in the strong-prior limit $\sigma \downarrow 0$. At the finite prior scale $\sigma^2 = N^{-\gamma}$, $\gamma \geq 1$, with no limit taken, the analogue of Theorem 1(c),(d) holds when $\theta < \theta_{L,\gamma}$ (E, Theorem 2.7): for competitors with $B \geq 1$ and n above a polynomial in $d, m, B, 1/\varepsilon, \log(1/\delta)$ (not the explicit n_0 above), with probability at least $1-\delta$ and at every width $N \geq m$, the expected clipped test loss of a posterior draw and the test loss of the posterior mean of the clipped outputs are at most $L_D(g) + \varepsilon$. Here $\theta_{L,\gamma} = 1/H$ when $L = 2$ or $\gamma \geq 2 - 1/H$; $\theta_{L,\gamma} = \max\{\frac{\gamma-1}{\gamma H - 1}, \frac{\gamma-1}{H(H-1)(2-\gamma)}\}$ when $1 < \gamma < 2 - 1/H$; and $\theta_{L,1} = 1/(4 + 16(L-2)3^{L-2})$ for $L \geq 3$, a case whose proof relies on an imported mean-field cover. E states no analogue of Corollary 2; for $H \geq 2$ and $\gamma \geq 2$ its explicit rate decays against $\mathcal{G}_n^{(\alpha)}$ only for $\alpha < 1/(8H)$ (E, Remark 2.8, in a comparison added after refereeing).
- **Close conditioning** (A, Theorem 4.4 and Corollary 4.5). With no or weak tempering, that is $c = 1$ or any c with $c\left(d^2 \log^2(d+1) + (dn)^{1/H}\right) = o(n)$, the learner fails at depth $L \geq 3$. On the noisy cube above with $n/2^d \to 1$, with probability at least $1 - e^{-2^{d/2}}$ and at learner widths above a polynomial in n, every near-minimizer, a posterior draw (with posterior probability tending to 1 as $\sigma \downarrow 0$) and the posterior means in the limit, clipped or not, have test loss at least $1 + \beta(1) - o(1)$ as $d \to \infty$, where $\beta(1) \approx 0.4848$, against 1 for the zero network. At finite d the certified excess is smaller (about 0.25 at $H = 2$, $d = 20$, for $c = 1$ and $n = 2^d$ with probability 0.99; A, Table 4.1).

Idea of the proof. As $\sigma \downarrow 0$ the posterior concentrates on near-minimizers of $F_{c,S}$. Comparing with a low-energy width-N copy of the competitor bounds their training loss by $\hat{L}_S(g) + (E_* + 1)\, n^{-\theta}$, and comparing with the zero network bounds their energy by $n^{\theta} + 1$. A Rademacher bound for tanh networks of bounded energy, valid at every width, then controls the gap between the training and test loss of the clipped outputs. It is a path-norm bound on the last H layers, with the first layer charged through the VC dimension $d+1$ of halfspaces rather than through its energy, which is why H and not L appears: at energy $K \approx n^{\theta}$ the gap is of order $K^{H/2}\Gamma_n/\sqrt{n}$, which vanishes exactly when $\theta < 1/H$. Finally, Hoeffding's inequality for the fixed competitor gives $\hat{L}_S(g) \leq L_D(g) + (U+1)^2\sqrt{\ln(2/\delta)/(2n)}$; a union bound over the two events, each of probability at least $1 - \delta/2$, and Jensen's inequality for the posterior mean give Theorem 1.

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